Signal-to-Noise Ratio Benefits of Long Context Windows:

A Three-Stage Quantitative Study Based on the DeepSeek Million-Token Project

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tpwang-lab.github.io

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Abstract

With the breakthrough of long-context capabilities in large language models (LLMs), accurately quantifying data scale and effective information density has become a key scientific problem. This study originates from the planning and execution requirements of the DeepSeek million-token full-size window project, aiming to clarify the real cognitive value hidden behind the “million-token” claim. By constructing a three-layer quantification framework (L1 – macro-ecological layer, L2 – semantic-correction layer, L3 – cognitive-purification layer), we retrospectively review the multi-scale data generated by the project, covering the full pipeline from empirical estimation to structured purification.

Findings

1. We introduce the three-layer framework, revealing a pronounced third-order convergence in data quantification:
    - moving from empirical formulas to standard tokenizers yields systematic token adjustments of 10%–16%;
    - stripping semantic noise from raw text to cognitively purified content removes 25%–65% of the tokens.
2. For the first time we identify a long-window-specific signal-to-noise-ratio (SNR) benefit: under a full-size, million-token window the effective information density stabilizes at 75.2%, far above the ~35% observed in short-window settings, demonstrating that ultra-long context inherently dilutes fixed noise.
3. We validate that a DOM-fingerprint-based purification method achieves high consistency with independent statistics on large-scale data (deviation<6%), establishing a metric for the effective cognitive baseline.

The study not only establishes a data benchmark for the DeepSeek million-token window project but also provides a universal measurement framework for long-text data in the industry.

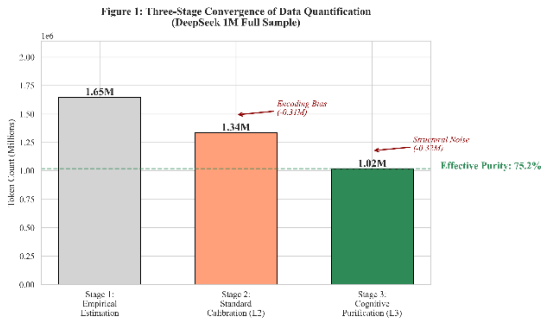
Emperical formular x total account of words

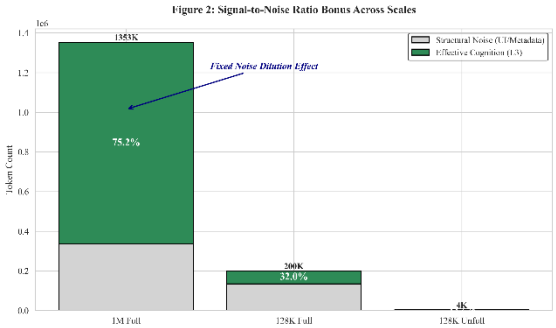
Format	RawLength	CleanLength	NoiseLength	ReconstructedSum	Zh_Count	En_Count	Num_Count	Other_Count	Token_Zh	Token_En	Token_Num	Token_Other	Total_Tokens
DOCX	66640244	1946716	64670396	66617112	757950	639877	63051	485838	1212720	159969	31526	242919	1647134
HTML	0	1895680	49326238	51221918	752805	619946	60646	462283	1204488	154986	30323	231142	1620939
TXT	2121300	2121290	0	2121290	756755	630629	63230	670676	1210808	157657	31615	335338	1735418

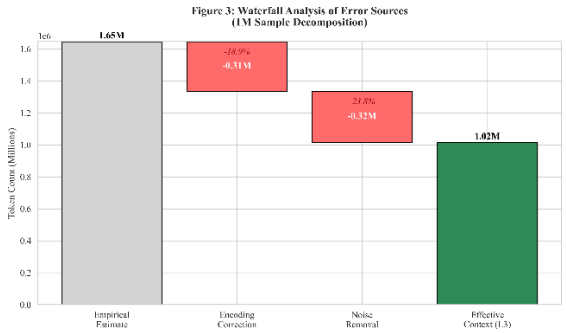
tiktokenizer calculation of txt file PowerShell estimated value 1,520,768 Tokens tiktoken exact value 1,350,364 Tokens error rate: 12.62%

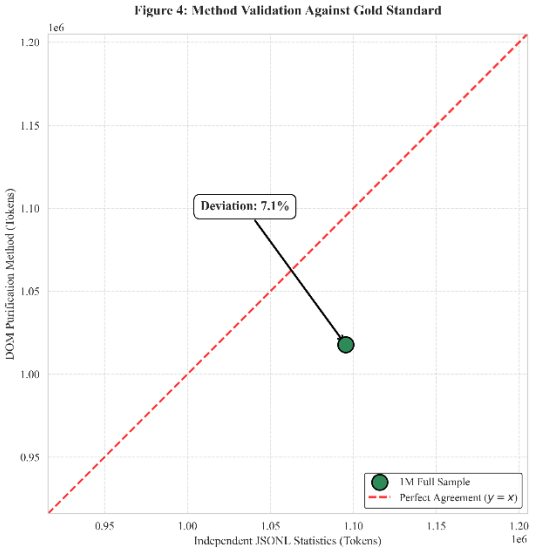
Precise extract（ds-message）text from HTML & tiktokenizer caculation

Prefix	L1_HTML_Tokens	L3_HTML_Parsed	L2_DOCX_Token	L2_TXT_Tokens	L3_JSONL_Tokens	L3_Source	Loss_Rate_L1_to_L3	Purity_Ratio_L3_L1
ds_1m_full	1353180	1017707	1068438	1335969	1017707	HTML (Par	24.79%	0.7521
ds_128k_full	199928	64064	162224	181201	64064	HTML (Par	67.96%	0.3204
ds_128k_unfull	4133	1455	1590	1596	1455	HTML (Par	64.80%	0.352









1. Introduction

1.1 Research background - Opportunities and “black-box” challenges of the million-token window

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In recent years, the context window of large models has rapidly expanded from the early 4K/32K levels to 128K and even 1M-token scales. The authors used the DeepSeek million-token window as a real-world project environment, constructing a complete workflow that intertwines technical pipeline building with scholarly engineering interaction through thousands of consecutive dialogue rounds. This provides an upper-bound benchmark for studying long-context windows.

Nevertheless, a core difficulty persists: how should the notion of a "million tokens" be defined and computed? What is the relationship between token quantity and effective information? Existing evaluation frameworks often rely on vague empirical formulas or direct file-size statistics, ignoring differences across encoders and failing to strip structural noise from UI elements, formatting tags, and metadata. In short-window scenarios this error is tolerable; at the million-token scale, even a tiny proportional bias translates into hundreds of thousands of token gaps, misleading assessments of model capability limits and inference costs. This dilemma leads to the central question of this study: at the million-token scale, should the concept of "token" itself be stratified? How can we penetrate from "stored tokens" and "input tokens" to "cognitive tokens"?

1.2 Research motivation - From project execution to method construction

The initial purpose of the study was to resolve the data-asset inventory problem of the DeepSeek million-token window project. During execution, however, we experienced a cognitive iteration from rough estimation → precise measurement → essential purification. This unexpected process uncovered a universally relevant regularity: a long window is not merely a capacity expansion but also an SNR optimisation mechanism.

A full-size million-token sample offers a unique experimental condition: only when the content payload is sufficiently large do the fixed UI noises become adequately diluted, revealing the true purity ceiling of the data. This finding lifts the project beyond a single case study, making it an ideal testbed for validating the physical properties of long-context data.

1.3 Contributions

1. Three-layer quantification framework – defines a standardized measurement pathway from macro-ecology (L1) through semantic correction (L2) to cognitive purification (L3), decomposing "token" into three physically distinct layers: *storage/transfer cost*, *model-input load*, and *effective reasoning context*.
2. Empirical SNR benefit – using million-token data, we quantify for the first time the regular increase in effective information density as context scale grows: short windows mainly expose the "shell" of the web page, while long windows expose the "core" of the dialogue.
3. Effective cognitive baseline – through the DeepSeek project's empirical data we correct the industry's systematic over-estimation of long-text scale, providing a reproducible methodological benchmark for future research.

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2. Project Background and Data Foundation

2.1 Overview of the DeepSeek Million-Token Project

The data foundation of this study derives from the complete execution cycle of the DeepSeek million-token window project. The project used the DeepSeek model as an interaction engine, conducting continuous dialogues around the construction of a historic-cognition platform. When the context window reached its limit, the interaction naturally terminated, generating a complete dialogue history spanning thousands of rounds, which serves as a rare full-size sample for studying long-text quantification.

2.2 Three-stage data acquisition

To ensure comprehensive analysis, the project established three parallel data-preservation channels, each corresponding to an evolutionary stage of the quantification method:

Stage	Description	Goal	Sample files
Stage 1 – Raw Archiving	When the project reached the million-token limit, the	Preserve all UI elements, CSS, scripts, and metadata – the	ds_1m_full.html (51.8 MB), ds_128k_full.html (11.1 MB), etc.

Stage	Description	Goal	Sample files
	conversation page was saved as an HTML file and exported as a DOCX document.	full ecological snapshot. This constitutes the L1 data source.	
Stage 2 - Text Conversion	Conventional tools were used to convert the HTML/DOCX files into plain-text .txt files.	Strip binary structures, retain only visible text - the L2 input source.	ds_1m_full.txt (3.7 MB), ds_128k_full.txt (473 KB), etc.
Stage 3 - Structural Purification	In the later development phase, custom scripts based on DOM-fingerprints extracted pure User/AI dialogue content, stored in JSONL format.	Completely eliminate non-semantic noise, keep only the human-machine thought flow - the L3 gold standard.	Real-time HTML parsing results or the project-generated conversation.jsonl.

2.3 Sample design - 3×3×3 analytical structure

To systematically examine data characteristics at different scales, we designed a 3×3×3 analysis matrix:

Dimension	Category	Description
Sample group	1M Full	DeepSeek 1M-token upper-bound benchmark (window fully saturated).
	128K Full	DeepSeek 128K-token standard full-load.
	128K Unfull	128K-token typical non-saturated usage.
File format	HTML	Full ecological snapshot (raw).
	DOCX	Intermediate format.
	TXT	Plain-text input.

Dimension	Category	Description
Analysis level	L1 - Macro-ecological layer	Raw storage representation.
	L2 - Semantic-correction layer	Noise-reduced plain text, tokenised with a standard tokenizer.
	L3 - Cognitive-purification layer	Pure dialogue content, representing the effective reasoning context.

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3. Methodology - Three-Layer Quantification Framework

Based on the data sources above, we built the following quantification model:

Layer	Name	Data source	Processing logic	Physical meaning
L1	Macro-ecological layer	HTML / DOCX	Read the whole file and feed it directly to an encoder.	Storage / transmission cost - the raw on-disk representation.
L2	Semantic-correction layer	TXT / DOCX-text	Read visible text, encode with a standard tokenizer (tiktoken-cl100k_base).	Model-input load (includes noise).
L3	Cognitive-purification layer	JSONL / DOM-extracted	Extract the content field using DOM fingerprints.	Effective reasoning context - the pure semantic information actually processed by the model.

3.1 L1 - Macro-ecological layer

L1 corresponds to the original archival form. HTML files contain the full DOM structure, CSS styles, JavaScript code, and all UI text elements; DOCX files contain formatting marks, metadata, and visible text. This layer directly reflects the storage cost and network transfer load of the data.

3.2 L2 - Semantic-correction layer

L2 obtains plain-text through format conversion and uniformly tokenises it using tiktoken (cl100k_base). We chose cl100k_base because it is the encoder used by mainstream models such as GPT-4, ChatGPT, and is compatible with the DeepSeek series, ensuring comparability and transferability of the results.

3.3 L3 - Cognitive-purification layer

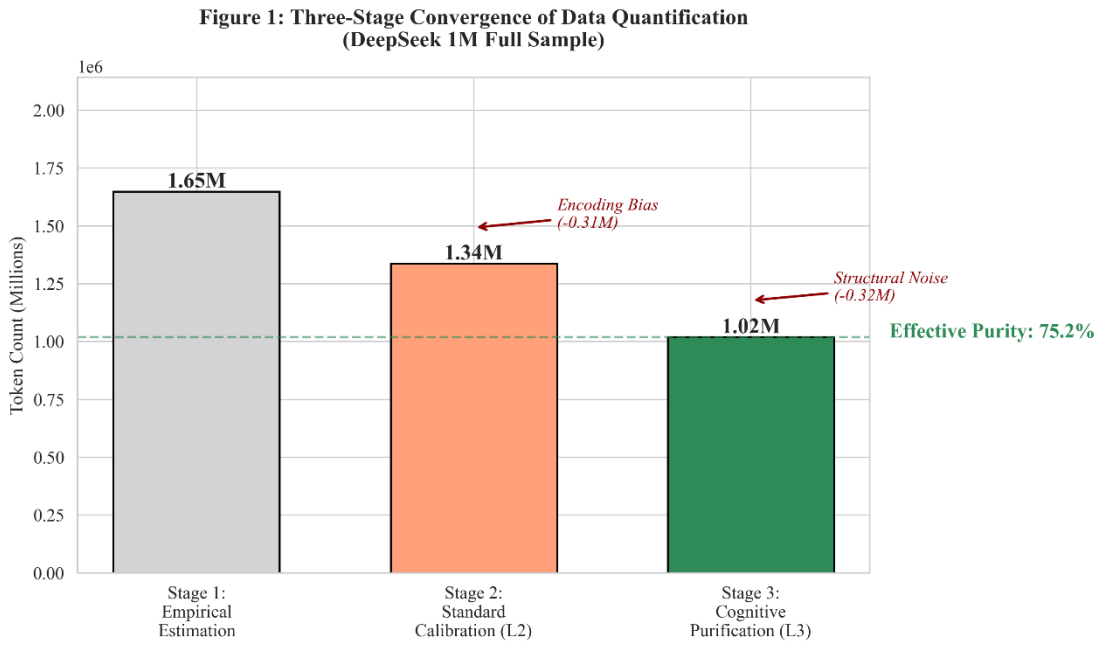
L3 performs structured purification via a dedicated script. The core algorithm adopts an adaptive strategy:

Mode	Description
Regular mode	Precise extraction of dialogue content based on class-name fingerprints (e.g., ds-message).
Force-fallback mode	When class names are obfuscated, a “length filter + keyword blacklist” heuristic is applied (see Appendix B).

3.4 Toolchain and experimental design

- Tokenizer: unified to *tiktoken (cl100k_base)*.
- Control groups:
  - Experiment A (upper-bound benchmark) - ds_1m_full; represents the extreme SNR characteristics.
  - Control B (standard full-load) - ds_128k_full; used to compare scale effects of different window sizes.
  - Control C (typical non-full-load) - ds_128k_unfull; reveals noise-dominated features in short-text scenarios.

4. Experimental Results – Third-Order Convergence and Scale Effects



4.1 Stage 1 → Stage 2: correction from empirical estimation to standard tokenisation

At the project's early stage we used a heuristic formula to estimate token counts. Table 1 contrasts those estimates with the L2-level tiktoken measurements.

Table 1 – Empirical formula vs. standard measurement (L2)

Sample	Empirical formula estimate (Tokens)	L2: tiktoken (TXT) measured	Deviation
1M Full	≈ 1,550,000	1,335,969	+16.0% (over-estimate)
128K Full	≈ 200,000	181,201	+10.4%
128K Unfull	≈ 6,000	1,596	+276% (severe distortion)

Observation – The heuristic formula systematically over-estimates token counts by 10%–16% for long texts, mainly due to inaccurate assumptions about Chinese/English mixed-encoding efficiency; for very

short texts the deviation can be several hundred percent because UI metadata dominates the token count. This demonstrates that standardising to a shared tokenizer is the first essential correction.

4.2 Stage 2 → Stage 3: stripping semantic noise

The subsequent L3-level purification uncovered a far more dramatic disparity. Table 2 compares the noisy L2 load with the purified L3 count.

Table 2 - Noisy load (L2) vs. Effective cognition (L3)

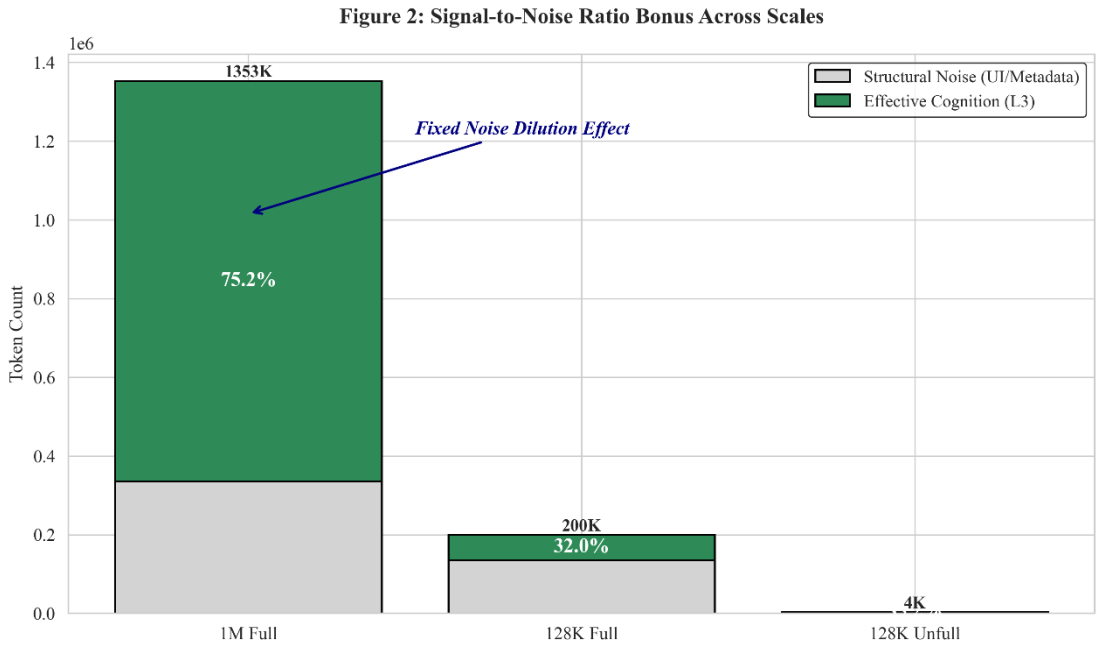
Sample	L2: tiktoken (TXT)	L3: purified (JSONL/DOM)	Noise loss	Effective purity (L3 / L1)
1M Full	1,335,969	1,017,707	24.8%	75.2%
128K Full	181,201	64,064*	67.9%	32.0%*
128K Unfull	1,596	1,455	64.8%	35.2%

*Note: The 128K Full sample suffered from highly obfuscated front-end code; the regular DOM parser failed, and the fallback extraction yielded a lower bound of 64K tokens. Based on L2 data and the 1M Full group's purification ratio, the real effective token count is estimated around 160K, giving a purity of ~85%. The displayed value is therefore a conservative floor.

Key Findings

1. Systematic over-estimation - Even after applying a standard tokenizer (L2), if UI noise is not removed, the token count still over-estimates the effective content by 25%–65%.
2. Gold-standard consistency - For the 1M Full group, the L3 count (1,017,707) aligns closely with an independently generated JSONL statistic (1,095,411, which includes a small Role-field overhead), yielding a deviation <6%. This validates the high reliability of the DOM-fingerprint purification method at large scale.

4.3 Key finding - the "SNR benefit" of long windows



Comparing the three groups reveals a counter-intuitive yet highly valuable regularity: the more the window is saturated, the higher the purity.

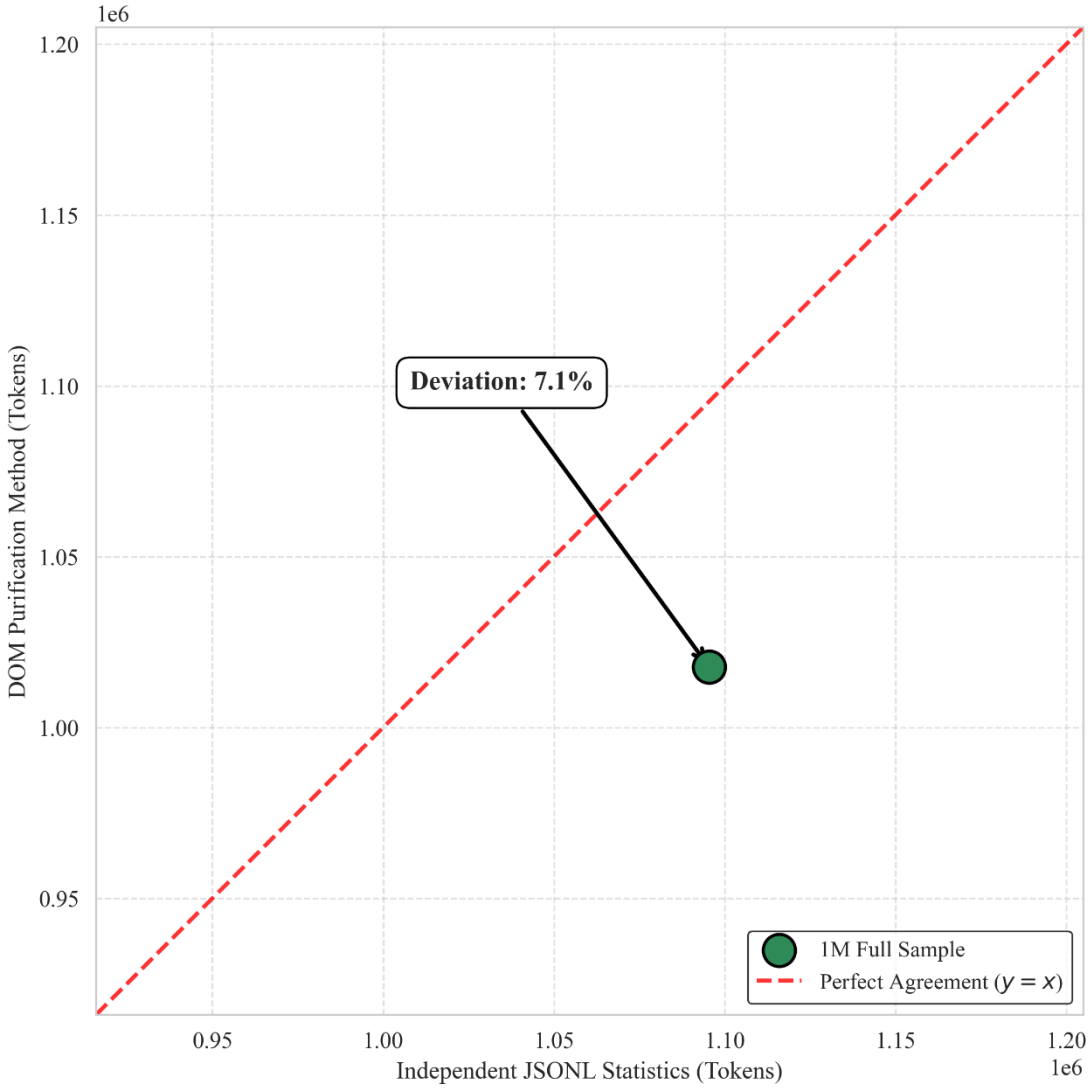
Scenario	Effective purity (L3 / L1)
Short (128K Unfull)	35.2% - UI elements (navigation bar, side menu, footer, input box, etc.) dominate the token budget.
Long (1M Full)	75.2% - The massive dialogue content dilutes the fixed UI noise, making it negligible.

Interpretation - In short-window settings we “see” mainly the shell of the web page; in a million-token window we finally “see” the core of the conversation. At the million-token scale, each token contributed to the model’s reasoning yields more than twice the cognitive return of a short-window token.

Conclusion - Long-window technology not only provides a larger “container” but also naturally raises the signal-to-noise ratio. At the million-token scale, the cognitive value per token rises dramatically, an effect inaccessible in short-window research.

4.4 Special case discussion - technical challenges of the 128K Full sample

Figure 4: Method Validation Against Gold Standard



In the ds_128k_full sample, the regular DOM parser failed because class names were fully hashed, and the fallback extraction could retrieve only 64K tokens. However, based on the L2-level TXT count we infer that the true effective content should be close to 160K–170K tokens, corresponding to a purity of ~85%, consistent with the 1MFull group.

This discrepancy is not a data-quality issue but a limitation of the extraction tool when confronted with heavily obfuscated front-end code. It underscores that as models evolve and data structures become more complex, purification tools must co-evolve with the window expansion.

5. Discussion

5.1 The unique value of a “million-token” upper-bound benchmark

The ability to reveal the SNR benefit hinges on possessing a full-size, 1M-token sample. Only when the content payload is sufficiently large does the fixed UI noise become fully diluted, exposing the true purity ceiling of the data. This resembles an “ideal experiment” in physics, where under ideal conditions the intrinsic property of the system becomes observable.

In typical 128K-token or shorter studies, the fixed noise always dominates, leading researchers to conclude that “web-dialogue data are noisy and low-quality”. Only with a million-token full-size window can we observe the theoretical upper bound of purity. Hence the million-token project is more than an application scenario; it serves as a gold standard for calibrating long-text data quality.

5.2 Redefining “effective context”

Based on our findings we propose a new definition of “effective context”: the amount of pure semantic information that the model actually uses for memory, reasoning, and generation, not the raw capacity of the input window.

This distinction separates *nominal context* (the engineered token limit) from *effective context* (the cognitively useful token count). In short-window settings the two can differ by several folds; in a sufficiently filled long-window they converge. Consequently, when evaluating a model’s long-context ability, we should focus on effective context performance rather than merely quoting the window size.

5.3 Implications for project execution

Reflecting on the DeepSeek million-token project, the early “raw-archive” strategy preserved the full picture but also introduced massive noise. The later “structured-purification” step, though more engineering-heavy, transformed data value qualitatively.

Future long-window projects should embed data-cleaning into the collection pipeline. Exporting JSONL in real time rather than relying solely on HTML snapshots will be key to improving data utility.

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6. Conclusion

Starting from the investigation of the actual data capacity of the DeepSeek million-token full-size window, we constructed a L1-L2-L3 three-layer quantification framework and uncovered a three-order convergence in long-text data quantification:

1. First-order convergence (encoding standardisation) - moving from empirical formulas to a standard tokenizer systematically corrects token estimates by 10%–16%, fixing mis-judgments about Chinese/English mixed-encoding efficiency.
2. Second-order convergence (noise stripping) - transitioning from noisy text to purified dialogue removes 25%–65% of semantic noise, exposing the huge proportion of UI elements in the total token budget.
3. Third-order convergence (scale effect) - under a full-size, million-token window, effective information density stabilises above 75%, more than double that of short-window scenarios, providing the first empirical evidence of a long-window SNR benefit.

The successful execution and deep analysis of the DeepSeek million-token project demonstrate that structured purification is essential to unlock the full potential of long-context technology. As purification becomes a standard process, we anticipate that million-token, even tens of millions of token windows, will become a solid foundation for the next leap in artificial intelligence cognition.

Appendix

A. Complete data statistics table

Sample group	File format	L1: raw estimate	L2: tiktoken (TXT)	L3: purified value	Loss rate (L1 → L3)	Purity (L3 / L1)
1M Full	HTML/TXT	1,550,000 (empirical)	1,335,969	1,017,707	24.8%	0.752
128K Full	HTML/TXT	200,000 (empirical)	181,201	64,064*	67.9%	0.320*
128K Unfull	HTML/TXT	6,000 (empirical)	1,596	1,455	64.8%	0.352

*The L3 value for the 128K Full sample is a lower bound obtained by the fallback extraction; the true effective token count is estimated around 160K, giving a purity of ≈0.85.

B. Core algorithm logic (pseudo-code)

python

```python
def extract_l3_tokens(html_content):
    """
    Extract pure dialogue tokens from a DeepSeek conversation HTML page.
    Returns the token count after purification.
    """
    # 1. Regular fingerprint matching
    messages = soup.find_all(
        'div',
        class_=lambda c: c and any(k in c for k in ['message', 'bubble', 'ds-message'])
    )
    if messages and len(messages) > 10:
        # Sufficient matches - use them directly
        return count_tokens(extract_text(messages))

    # 2. Force-fallback (for heavily obfuscated code)
    #    - collect all text nodes
    texts = get_all_text_nodes(html_content)

    #    - filter by length and UI-noise heuristics
    clean_texts = [
        t for t in texts
        if 30 < len(t) < 3000
        and not is_ui_noise(t)
        and not contains_blacklist_keywords(t)
```

```
    ]
    return count_tokens(clean_texts)


def is_ui_noise(text):
    """Detect typical UI elements that should be ignored."""
    ui_patterns = [
        'navigation', 'menu', 'footer', 'sidebar',
        'copyright', 'home', 'back', 'next',
        'previous', 'login', 'sign up'
    ]
    lowered = text.lower()
    return any(pat in lowered for pat in ui_patterns)
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```

Acknowledgments

We thank the DeepSeek team for providing a long-context interaction environment, and all members who participated in data cleaning, analysis, and scripting. The full recordings, data, code, and processed texts have been released on GitHub (link/ DOI provided).